

Yunhan Qiao

Ph.D. Candidate, Computer Science · Oregon State University
JohnsonYqiao@gmail.com | +1 480-791-7396

Research Profile

I am a Ph.D. candidate in Computer Science researching empirical software engineering, code comprehension, and human–AI collaboration. My work designs and evaluates developer-facing AI systems through controlled experiments, behavioral analysis, and mixed-methods research. Through studies of how students and expert developers adopt GenAI assistants such as GitHub Copilot and Cursor in realistic brownfield programming tasks, I investigate how AI assistance reshapes code comprehension, cognitive effort, and task performance—with the goal of informing the design of future AI coding agents and IDEs. I also maintain a strong interest in full-stack software development, AI agent development, and UX design.

Education

Oregon State University

Sep 2023 – Jun 2027 (expected)

Ph.D. in Computer Science — Corvallis, OR

Advisor: Dr. Christopher Hundhausen

Research Interests: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction

Arizona State University

Aug 2021 – May 2023

M.S. in Software Engineering — Tempe, AZ

Advisor: Dr. Robert LiKamWa

Thesis: Networked System for Volumetric Athletic Coaching in Augmented Reality

Guangxi University of Science and Technology & Southern Cross University

Sep 2016 – Jun 2020

Dual Degree Program — Liuzhou, China

B.E. in Software Engineering & B.E. in Information Technology

Research Topics

GitHub Copilot's Effects on Computing Students in Brownfield Programming Tasks Mar 2024 – Jun 2025

- Designed a controlled experimental protocol using GitHub Classroom to evaluate GenAI-assisted code comprehension in brownfield programming tasks using a 3K-LOC plain JavaScript codebase.
- Developed a multidimensional comprehension assessment framework measuring system objective, implementation, debugging, reverse engineering, and open-ended reasoning.
- Quantified the impact of GenAI assistance on 15 computing students, finding a 50.7% reduction in task completion time and a 71.4% increase in task correctness.
- Identified a comprehension–performance tradeoff, finding that active verification of GenAI-generated code correlated with substantially stronger code comprehension.

How Expert Developers Use GitHub Copilot in Brownfield Programming Tasks Jul 2025 – Present

- Designed and conducted a mixed-methods study investigating how expert developers adopt GenAI during brownfield programming tasks within an 88K-LOC MERN stack (MongoDB, Express, React, Node) codebase.
- Collected and analyzed over 30 hours of programming sessions with 10 expert developers through AWS WorkSpaces.
- Developed a behavioral analysis framework combining Information Foraging Theory and state machines to synthesize GenAI-assisted comprehension and problem-solving strategies.

- Identified patterns of AI reliance, verification, and information seeking that influence both task quality and codebase understanding, informing the design of future developer AI systems.

Brain Activity of Vibe Coding on Developers

Feb 2026 – Present

- Co-designed a controlled EEG study exploring the cognitive effects of GenAI assistants (Cursor and GitHub Copilot) in brownfield programming tasks using an 88K-LOC MERN stack codebase.
- Conducted in-person empirical studies with 30 developers, collecting neurophysiological data during vibe coding.
- Applied Information Foraging Theory to model shifts in information-seeking behavior under Ask and Agent modes of GenAI.
- Investigated correlations among code comprehension, cognitive effort, and task performance to derive evidence-based design principles for future AI coding agents/IDEs.

Other Research Topics

LLM-Guided Differential Fuzzing for Detecting Heterogeneous Bugs

Jul 2023 – Sep 2024

- Developed a Python-based LLM-guided seed input generator.
- Ran experiments using Bash on a High Performance Computing (HPC) cluster and successfully detected 8 of 20 heterogeneous bugs for LAMMPS, a widely used open-source HPC application.

Systematic Literature Review (SLR) of GenAI for Code Comprehension

Sep 2024 – Jun 2025

- Systematically analyzed 31 research papers using thematic analysis to study how GenAI tools impact developer workflows and codebase comprehension.
- Synthesized empirical data into five actionable GenAI approaches (e.g., software explanation and readability enhancement) while benchmarking their effectiveness and analysis methodologies.

Reflection and Equitable Participation in Team Software Projects

Feb 2025 – Oct 2025

- Conducted qualitative analysis to evaluate the efficacy of a Slackbot intervention designed to improve contribution equity in software development teams.

Publications

Published / Accepted

Y. Qiao, M. I. H. Shihab, & C. Hundhausen. A Systematic Literature Review of the Use of GenAI Assistants for Code Comprehension: Implications for Computing Education Research and Practice. *ACM Transactions on Computing Education (TOCE)*, 26(2), 1–33, 2026.

M. I. H. Shihab, C. Hundhausen, A. Tariq, S. Haque, **Y. Qiao**, & B. W. Mulanda. The Effects of GitHub Copilot on Computing Students' Programming Effectiveness, Efficiency, and Processes in Brownfield Coding Tasks. *ACM Conference on International Computing Education Research (ICER)*, Vol. 1, 407–420, 2025.

A. Tariq, P. Conrad, C. Hundhausen, S. Chandrasekaran, **Y. Qiao**, & S. Haque. Using Slackbot-Prompted Reflections to Promote Participation and Reduce Contribution Inequalities in Team Software Development Projects. *ACM Transactions on Computing Education (TOCE)*, 2026.

M. Motwani, A. Kulkarni, **Y. Qiao**, M. Davis, & Z. Chen. LLM-Guided Differential Fuzzing for Detecting Platform-Specific Bugs in Scientific Applications. *IEEE International Conference on AI x Software Engineering (AIxSE)*, 2025.

Under Submission

Y. Qiao, M. I. H. Shihab, S. Haque, & C. Hundhausen. Code Comprehension with GitHub Copilot: Performance Gains, Comprehension Trade-offs, and Behavioral Predictors in Brownfield Programming. *ACM Technical Symposium on Computer Science Education (SIGCSE)*, 2027.

Y. Qiao, M. I. H. Shihab, A. Tariq, & C. Hundhausen. Unraveling the Codebase: Exploring GenAI Adoption for Program Comprehension Among Expert Developers in Brownfield Programming Tasks. *ACM Conference on Human Factors in Computing Systems (CHI)*, 2027.

Y. Qiao, et al. Brain Activity During Vibe Coding: An EEG Study of Cognitive Load and Flow. *ACM Conference on Human Factors in Computing Systems (CHI)*, 2027.

Teaching Experience

Graduate Teaching Assistant — Oregon State University

2024 – Present

- CS 362: Software Engineering II
- CS 492: Mobile Software Development
- CS 467: Online Capstone Project

Skills and Coursework

Programming	Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
Frameworks & Tools	React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Qualtrics
Artificial Intelligence	Machine Learning; NLP with Deep Learning; Software Engineering for Machine Learning
Software Engineering	Program Analysis and Evaluation; Algorithms and Data Structures; Foundations of Software Engineering
HCI	Human–Computer Interaction; Field Studies in SE and HCI
Statistics	Methods of Data Analysis
Mathematics	Calculus; Linear Algebra; Probability

Service

Program Committee: SIGCSE TS, 2027

Languages

English (Fluent), Mandarin Chinese (Native)